Ordering fractions by way of a common denominator

Task A: Review writing equivalent fractions

1) Fill in the missing numerator or denominator.

a) $\frac{9}{11} = \frac{90}{110}$

b) $\frac{3}{5} = \frac{15}{25}$

c) $\frac{28}{32} = \frac{7}{8}$

d) $\frac{24}{36} = \frac{2}{3}$

e) $\frac{5}{8} = \frac{40}{64}$

f) $\frac{11}{15} = \frac{44}{60}$

g) $\frac{91}{156} = \frac{7}{12}$

h) $\frac{340}{51} = \frac{20}{3}$

2) Which of the following are true?

a) $\frac{2}{3} = \frac{30}{48}$ $\frac{\times 15}{\times 16}$ False

b) $\frac{4}{9} = \frac{16}{36}$ $\frac{\times 4}{\times 4}$ True

c) $\frac{7}{8} = \frac{21}{24}$ $\frac{\times 3}{\times 3}$ True

d) $\frac{8}{11} = \frac{56}{66}$ $\frac{\times 7}{\times 6}$ False

e) $\frac{96}{144} = \frac{2}{3}$ $\frac{\div 48}{\div 48}$ True

f) $\frac{256}{75} = \frac{16}{5}$ $\frac{\div 16}{\div 15}$ False

Task B: Comparing fractions

1) Use equivalent fractions to decide which fraction is greater.

Insert either < or > to make the statement true.

a) $\frac{3}{8} \boxed{<} \frac{10}{24}$

b) $\frac{4}{7} \boxed{>} \frac{24}{49}$

c) $\frac{8}{21} \boxed{>} \frac{5}{14}$

d) $\frac{7}{11} \boxed{<} \frac{2}{3}$

e) $\frac{5}{9} \boxed{<} \frac{5}{6}$

f) $\frac{5}{12} \boxed{<} \frac{7}{15}$

g) $\frac{1}{7} \boxed{>} \frac{3}{31}$

h) $\frac{25}{75} \boxed{>} \frac{589}{1800}$

2)

a) Place the words ‘greater’ and ‘less’ in the sentence:

$\frac{3}{11}$ is *less* than $\frac{4}{11}$ but *greater* than $\frac{2}{11}$

b) Jacob says “ $\frac{3}{11}$ is exactly halfway between $\frac{4}{11}$ and $\frac{2}{11}$.”

Is he correct?

Jacob is correct as there is $\frac{1}{11}$ *between* $\frac{3}{11}$ *and* $\frac{4}{11}$, *as well as between* $\frac{3}{11}$ *and* $\frac{2}{11}$.

Task C: Order fractions using equivalent fractions

Write the following fractions with a common denominator and order from smallest to largest.

a) $\frac{7}{10}$ $\frac{3}{4}$ $\frac{3}{5}$ $\frac{11}{20}$

$\boxed{\frac{11}{20} \quad \frac{3}{5} \quad \frac{7}{10} \quad \frac{3}{4}}$

b) $\frac{3}{4}$ $\frac{9}{16}$ $\frac{5}{8}$ $\frac{1}{2}$

$\boxed{\frac{1}{2} \quad \frac{9}{16} \quad \frac{5}{8} \quad \frac{3}{4}}$

c) $\frac{13}{20}$ $\frac{17}{25}$ $\frac{7}{10}$ $\frac{3}{5}$ $\frac{3}{4}$

$\boxed{\frac{3}{5} \quad \frac{13}{20} \quad \frac{17}{25} \quad \frac{7}{10} \quad \frac{3}{4}}$

d) $\frac{4}{5}$ $\frac{2}{3}$ $\frac{9}{12}$ $\frac{5}{6}$ $\frac{11}{15}$

$\boxed{\frac{2}{3} \quad \frac{11}{15} \quad \frac{9}{12} \quad \frac{4}{5} \quad \frac{5}{6}}$